QUESTION 1:

What is a Common Table Expression (CTE), and how does it improve SQL query readability?

➢ A CTE is a temporary result set that you can reference within a ‘`SELECT`’, ‘`INSERT`’, ‘`UPDATE`’ or ‘`DELETE`’ statement. It is defined using the ‘`WITH`’ keyword. It functions like a virtual table that exists only for the duration of a single query, improving the readability and organization of complex SQL code.

It improves readability by:

- **Modularization:** It breaks complex, nested subqueries into logical "blocks."
- **Sequential Logic:** It allows us to write queries in the order they are logically processed (Step A, then Step B), rather than inside-out like subqueries.
- **Self-Documentation:** we can give the CTE a meaningful name, making it clear what that specific part of the data represents.

QUESTION 2:

Why are some views updatable while others are read-only? Explain with an example.

➢ A view is updatable only if the database can map the change back to a single, unambiguous row in the underlying base table.

- **Updatable Views:** These are typically simple views that select columns from one table without any complex transformations.
  - Example: A view selecting Name and Email from a Users table. Updating a name in the view clearly updates that specific user's row.
- **Read-Only Views:** These occur when the view contains elements that "mask" the source data, such as Aggregations (SUM, AVG), JOINS between multiple tables, or the DISTINCT keyword.
  - Example: A view showing Total_Sales per Region. If you try to update the "Total Sales" figure, the database doesn't know which individual transaction row to change to reach that new total.

QUESTION 3:
What advantages do stored procedures offer compared to writing raw SQL queries repeatedly?

- **Performance:** They are compiled once and stored in the database. Subsequent executions are faster because the execution plan is already generated.
- **Reduced Network Traffic:** Instead of sending a long SQL script over the network, you only send the procedure name and parameters.
- **Security:** Users can be given permission to execute a procedure without having direct access to the underlying tables.

- **Code Reusability:** You write the logic once and call it from any application or interface, ensuring consistency.

QUESTION 4:
What is the purpose of triggers in a database? Mention one use case where a trigger is essential.

➢ The purpose of a trigger is to automatically execute a piece of code in response to a specific event (like INSERT, UPDATE, or DELETE) on a table. They are used to maintain data integrity and automate background tasks.

Essential Use Case: Audit Logging. If you need to track who changed a salary record and when it happened, a trigger can automatically capture the old value, the new value, and the timestamp, and write it to an Audit_Log table every time an update occurs.

QUESTION 5:
Explain the need for data modelling and normalization when designing a database.

Data modeling and normalization are the foundations of a healthy database.

- **Data Modelling:** This is the blueprinting phase. It helps stakeholders understand how data entities (like Customers, Orders, Products) relate to each other, ensuring the database meets business requirements.
- **Normalization:** This is the process of organizing data to reduce redundancy and dependency**.**
    - **Need for Normalization:**
        1. Prevents Inconsistencies: If a customer's address is stored in five different places and they move, you might forget to update one, leading to "dirty data."
        2. Saves Storage: By not repeating the same data across thousands of rows.
        3. Data Integrity: It ensures that data is logically stored (e.g., you can't have an order for a product that doesn't exist).